

CH#13Uncertainty

Things don't happen as you expect
There's always a chance that



Contents

- Uncertainty
- Probability
- Syntax and Semantics
- Inference
- Independence and Bayes' Rule

Uncertainty

The cost of choosing

I need to be at the airport at 9:00 pm

- ▲ Agent 1: 95% chance of making it departing at 7:00pm
- ▲ Agent 2: 98% chance of making it departing at 8:00pm
- ▲ Agent 3: 100% chance of making it departing at 10:00pm

Which one do you choose? Why?

Agent 2: 98% chance of making it departing at 8:00pm

Uncertainty

Example

What would need to happen in the KB to diagnose a cavity.

- △ *Toothache* $\Rightarrow$ *Cavity*
- △ *Toothache* $\Rightarrow$ *Cavity* $\vee$ *GumProblem* $\vee$ *Abscess* . . .
- △ *Cavity* $\Rightarrow$ *Toothache*

But these can't be true or complete...

Probabilities are a way to summarize uncertainty

- △ With the facts I can observe ...
- △ There's 80% of chance of having a cavity

Rational decisions involve:

- △ Likelihood of achieving a state/goal (probability)
- △ Relative importance of the states/goals (Utility)

Decision Theory = probability theory + utility theory

Maximum Expected Utility: An agent is rational iff it chooses the action that yields the highest expected utility, averaged over all the possible outcomes of the action.

Probabilities in AI

Arguments Against²

- △ Prob. require massive amount of data
- △ Use of prob. require enumeration of all possibilities
- △ Hides details of character of uncertainty
- △ People are bad prob. estimators
- △ We do not have those numbers
- △ We find their use inconvenient

²Taken from V. Lesser's CS683

Introducing Probabilities in AI

Probability= It is a measure of how likely the event is to happen

It lies between 0-1

Probability

Likelihood Line



It will
Snow in winter

The answer to
 $2 + 7 = 9$

A year when
it does not
rain

Toss a coin
That land
Heads

Going without
Food and drink
for a year.

Probabilities

language of propositions

- △ **Random Variable:** variables like die_1 , $cavity$, *etc.*
- △ **Domain:** The set of possible values for a variable. e.g.
 $die_1 = \{1, 2, 3, 4, 5, 6\}$

What's the domain of *Weather*, *Cavity*, Age^3

Probabilities

- A **random variable** can take on one of a set of different values, each with an associated probability. Its value at a particular time is **subject to random variation**.
 - **Discrete** random variables take on one of a discrete (often finite) range of values
 - Domain values must be **exhaustive** and **mutually exclusive**
- For us, random variables will have a discrete, countable (usually finite) domain of **arbitrary values**.
 - Mathematical statistics usually calls these **random elements**
 - Example: Weather is a **discrete random variable** with domain {sunny, rain, cloudy, snow}.
 - Example: A **Boolean random variable** has the domain {true,false},

Probability Distribution Language

Let's consider $Weather = \{sunny, rainy, cloudy, snow\}$

△ $P(sunny) = 0.6; P(rain) = 0.1; P(cloudy) = 0.29; P(snow) = 0.01$

△ $P(Weather)$ is a **probability distribution**.

△ $P(Weather) =$

sunny	rainy	cloudy	snow
0.6	0.1	0.29	0.01

△ $P(Weather) = \langle 0.6, 0.1, 0.29, 0.01 \rangle$ for short

Probabilities

Concepts

- △ **Sample Space:** All possible worlds
 - △ dice roll: $\Omega = (1, 6)$
 - △ Ω = all possible worlds; ω = one world
- △ $0 \leq P(\omega) \leq 1$ for every ω
- △ $\sum_{\omega \in \Omega} P(\omega) = 1$

Probability = No. of Favorable events / Total no. of events

What's the possibility of obtaining a head in a coin toss?

1/2

Possibility of odd numbers in Die?

e.g., $P(1)=P(2)=P(3)=P(4)=P(5)=P(6)=1/6$.

Probabilities

More concepts and one example

- △ **Unconditional/Prior Probabilities:** $P(\text{dice are even})$ or $P(\text{doubles})$
- △ **Evidence:** Observed events. e.g. first die is 3.
- △ **Conditional/Posterior probability:** given priors and evidence we compute the posterior.
 - △ e.g. Prob of doubles given that first die is 3.
 - △ $P(\text{doubles} | \text{die}_1 = 3)$
 - △ $P(\text{cavity} | \text{toothache} = T)$

Joint probability distribution

- Probability assignment to all combinations of values of random variables (i.e. all elementary events)

	toothache	$\neg$ toothache
cavity	0.04	0.06
$\neg$ cavity	0.01	0.89

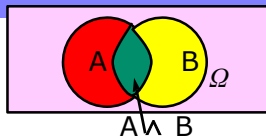


- The sum of the entries in this table has to be 1
- *Every question about a domain can be answered by the joint distribution*
- Probability of a proposition is the sum of the probabilities of elementary events in which it holds
 - $P(\text{cavity}) = 0.1$ [marginal of row 1]
 - $P(\text{toothache}) = 0.05$ [marginal of toothache column]



Conditional Probability

	toothache	$\neg$ toothache
cavity	0.04	0.06
$\neg$ cavity	0.01	0.89



- $P(\text{cavity})=0.1$ and $P(\text{cavity} \wedge \text{toothache})=0.04$ are both *prior* (unconditional) probabilities
- Once the agent has new evidence concerning a *previously unknown* random variable, e.g. Toothache, we can specify *a posterior* (conditional) probability e.g. $P(\text{cavity} \mid \text{Toothache}=\text{true})$

$$P(a \mid b) = P(a \wedge b) / P(b)$$

Where $P(b) > 0$

[Probability of a with the Universe Ω restricted to b]

→ The new information restricts the set of possible worlds ω_i consistent with it, so **changes the probability**.

Conditional Probability (continued)

- **Definition of Conditional Probability:**

$$P(a \mid b) = P(a \wedge b) / P(b)$$

- **Product rule gives an alternative formulation:**

$$\begin{aligned} P(a \wedge b) &= P(a \mid b) * P(b) \\ &= P(b \mid a) * P(a) \end{aligned}$$

- **A general version holds for whole distributions:**

$$P(\text{Weather}, \text{Cavity}) = P(\text{Weather} \mid \text{Cavity}) * P(\text{Cavity})$$

Probabilistic Inference

- **Probabilistic inference:** the computation
 - from *observed evidence*
 - of *posterior probabilities*
 - for *query propositions*.
- We use the **full joint distribution** as the “knowledge base” from which answers to questions may be derived.
- Ex: three Boolean variables *Toothache (T)*, *Cavity (C)*, *ShowsOnXRay (X)*

	t		$\neg t$	
	x	$\neg x$	x	$\neg x$
c	0.108	0.012	0.072	0.008
$\neg c$	0.016	0.064	0.144	0.576

- Probabilities in joint distribution sum to 1

Probabilistic Inference II

	t		$\neg t$	
	x	$\neg x$	x	$\neg x$
c	0.108	0.012	0.072	0.008
$\neg c$	0.016	0.064	0.144	0.576

- Probability of any proposition computed by finding atomic events where proposition is true and adding their probabilities
 - $P(\text{cavity} \vee \text{toothache})$
 $= 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064$
 $= 0.28$
 - $P(\text{cavity})$
 $= 0.108 + 0.012 + 0.072 + 0.008$
 $= 0.2$
- $P(\text{cavity})$ is called a marginal probability and the process of computing this is called marginalization

Probabilistic Inference III

	t		$\neg t$	
	x	$\neg x$	x	$\neg x$
c	0.108	0.012	0.072	0.008
$\neg c$	0.016	0.064	0.144	0.576

- Can also compute conditional probabilities.
- $P(\neg \text{cavity} \mid \text{toothache})$
= $P(\neg \text{cavity} \wedge \text{toothache}) / P(\text{toothache})$
= $(0.016 + 0.064) / (0.108 + 0.012 + 0.016 + 0.064)$
= 0.4
- Denominator is viewed as a *normalization constant*:
 - Stays constant no matter what the value of Cavity is.
(Book uses α to denote normalization constant $1/P(X)$, for random variable X .)

Bayes' Rule & Diagnosis

$$\text{Posterior} \quad P(a|b) = \frac{\text{Likelihood} \quad P(b|a) * \text{Prior} \quad P(a)}{\text{Normalization} \quad P(b)}$$

- Useful for assessing diagnostic probability from causal probability:

$$P(\text{Cause}|\text{Effect}) = \frac{P(\text{Effect}|\text{Cause}) * P(\text{Cause})}{P(\text{Effect})}$$

Bayes' Rule For Diagnosis II

$$P(\text{Disease} \mid \text{Symptom}) = \frac{P(\text{Symptom} \mid \text{Disease})}{P(\text{Symptom})} *$$

Imagine:

- disease = TB, symptom = coughing
- $P(\text{disease} \mid \text{symptom})$ is different in TB-indicated country vs. USA
- $P(\text{symptom} \mid \text{disease})$ should be the same
 - It is more widely useful to learn $P(\text{symptom} \mid \text{disease})$
- What about $P(\text{symptom})$?
 - Use *conditioning* (next slide)
 - For determining, e.g., the *most likely* disease given the symptom, we can just ignore $P(\text{symptom})$!!! (see slide 35)

Conditioning

- **Idea:** Use *conditional probabilities* instead of joint probabilities

- $$P(a) = P(a \wedge b) + P(a \wedge \neg b)$$

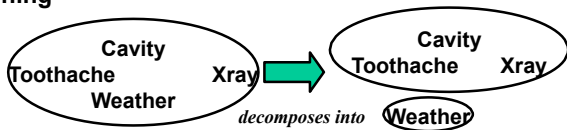
Here:
$$= P(a | b) * P(b) + P(a | \neg b) * P(\neg b)$$

$$P(\text{symptom}) = P(\text{symptom} | \text{disease}) * P(\text{disease}) + P(\text{symptom} | \neg \text{disease}) * P(\neg \text{disease})$$

- More generally: $P(Y) = \sum_z P(Y|z) * P(z)$
- Marginalization and conditioning are useful rules for derivations involving probability expressions.

The Solution: *Independence*

- Random variables **A** and **B** are independent iff
 - $P(A \wedge B) = P(A) * P(B)$
 - equivalently: $P(A | B) = P(A)$ and $P(B | A) = P(B)$
- A and B are independent if knowing whether A occurred gives no information about B (and vice versa)***
- Independence assumptions are *essential* for efficient probabilistic reasoning



$$P(T, X, C, W) = P(T, X, C) * P(W)$$



- 15 entries ($2^4 - 1$) reduced to 8 ($2^3 - 1 + 2 - 1$)
For n independent biased coins, $O(2^n)$ entries $\rightarrow O(n)$

Conditional Independence

- BUT **absolute** independence is rare
- Dentistry is a large field with hundreds of variables, none of which are independent. What to do?
- A and B are conditionally independent given C iff
 - $P(A \mid B, C) = P(A \mid C)$
 - $P(B \mid A, C) = P(B \mid C)$
 - $P(A \wedge B \mid C) = P(A \mid C) * P(B \mid C)$
- Toothache (T), Spot in Xray (X), Cavity (C)
 - None of these are independent of the other two
 - But **T and X are conditionally independent given C**

